LCD / LCM SPECIFICATION



WINSTAR Display Co.,Ltd.

華凌光電股份有限公司

SPECIFICATION

MODULE NO.: WG240128A

General Specification

Item	Dimension	Unit
Number of dots	240 x 128	—
Module dimension	170.0 x 93.4 x 12.4(MAX)	mm
View area	128.0 x 74.0	mm
Active area	119.97 x 63.97	mm
Dot size	0.47 x 0.47	mm
Dot pitch	0.50 x 0.50	mm
Duty	1/128	
Backlight Type	LED	
IC	RA6963	
Interface	8080	

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Absolute Maximum Ratings

Item	Symbol	Min	Typ	Max	Unit
Operating Temperature	T_{OP}	-20	–	+70	℃
Storage Temperature	T_{ST}	-30	–	+80	℃
Input Voltage	V_{IN}	-0.3	–	V_{DD}+0.3	V
Supply Voltage For Logic	V_{DD}-V_{SS}	-0.3	–	+7.0	V

Electrical Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
Supply Voltage For Logic	V_{DD}-V_{SS}	–	4.5	5.0	5.5	V
Supply Voltage For LCD	V_{DD}-V_0	Ta=-20℃	–	–	20.1	V
		Ta=25℃	17.5	18.0	18.5	V
		Ta=70℃	16.3	–	–	V
Input High Volt.	V_{IH}	–	0.8V_{DD}	–	V_{DD}	V
Input Low Volt.	V_{IL}	–	0	–	0.15 V_{DD}	V
Output High Volt.	V_{OH}	–	V_{DD}-0.3	–	V_{DD}	V
Output Low Volt.	V_{OL}	–	0	–	0.3	V
Supply Current	I_{DD}	–	15	30	60	mA

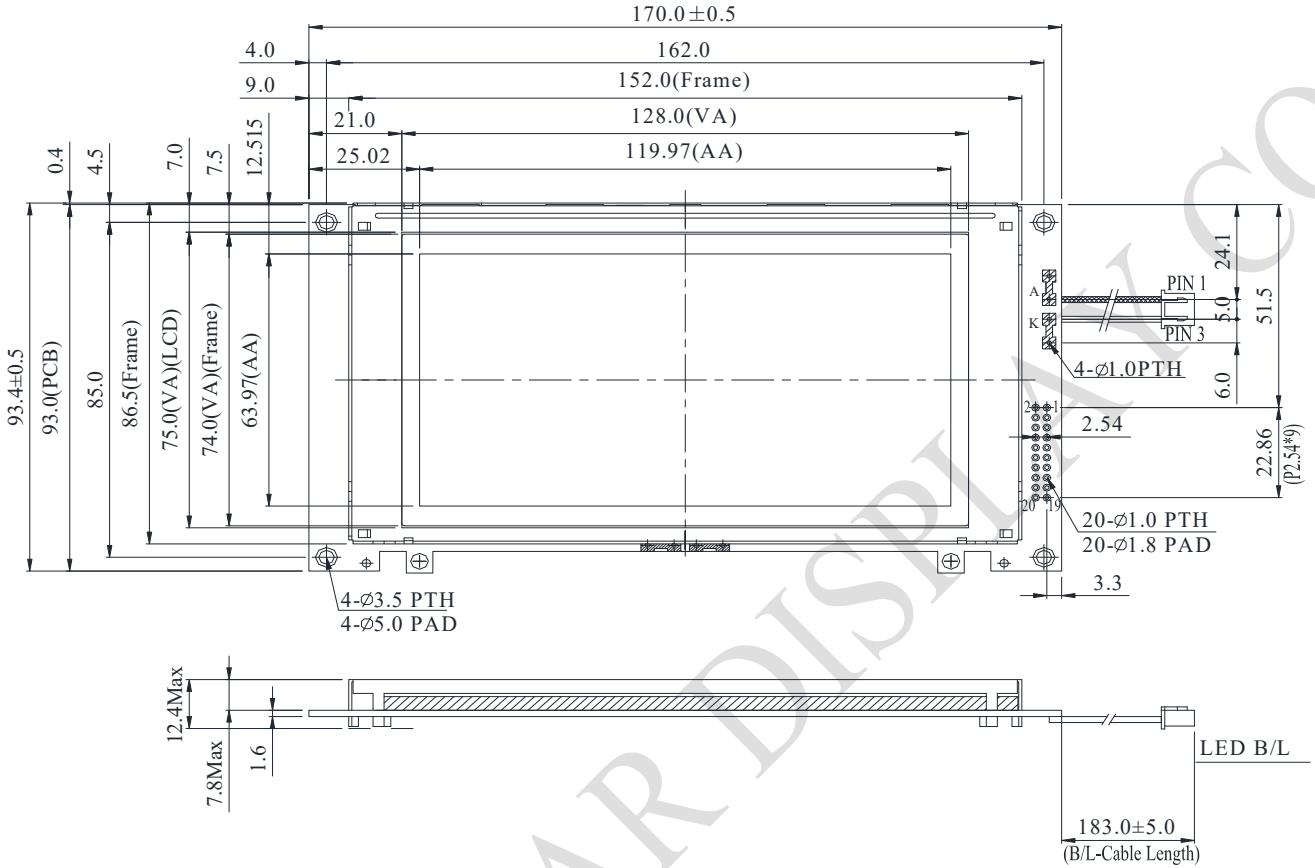
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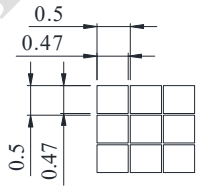
Interface Pin Function

Pin No.	Symbol	Level	Description
1	FGND	-	Frame ground (Connected to bezel)
2	Vss	-	GND
3	Vdd	-	Power supply
4	Vo	-	Power supply for LCD driver
5	WR	L	Data write. Write data into RA6963when WR = L
6	RD	L	Data read. Read data from RA6963when RD = L
7	CE	L	L : Chip enable
8	C/D	H / L	WR=L , C/D=H : Command Write C/D=L: Data write RD=L , C/D=H : Status Read C/D=L: Data read
9	Vee	-	Negative Voltage Output
10	RESET	H / L	H : Normal ; L : Initialize RA6963
11	DB0	H / L	Data bus line
12	DB1	H / L	Data bus line
13	DB2	H / L	Data bus line
14	DB3	H / L	Data bus line
15	DB4	H / L	Data bus line
16	DB5	H / L	Data bus line
17	DB6	H / L	Data bus line
18	DB7	H / L	Data bus line
19	FS	MD2	Pins for selection of font; H : 6 * 8 , L : 8 * 8
20	RV	H / L	H: Reverse L: Normal

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Contour Drawing





DOT SIZE
SCALE 10/1

PIN NO.	SYMBOL
1	FGND
2	Vss
3	Vdd
4	Vo
5	$\overline{WR}$
6	RD
7	$\overline{CE}$
8	C/D
9	Vee
10	$\overline{RESET}$
11	DB0
12	DB1
13	DB2
14	DB3
15	DB4
16	DB5
17	DB6
18	DB7
19	FS
20	RV

The non-specified tolerance of dimension is ±0.3mm.

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